Section A: Pure Mathematics [40 marks]

1 A sequence $x_1, x_2, x_3, \ldots$ is defined by

$$x_1 = a \text{ and } x_{n+1} = x_n + \left(\frac{3}{2}\right)^n,$$

where a is a constant.

(a) Find x_2, x_3 and x_4 in terms of a. [2]

(b) Find x_n in the form $p + q\left(\frac{3}{2}\right)^{n-1}$, where p and q are constants in terms of a, where appropriate. [2]

(c) Determine if the sequence $\{x_n\}$ converges. [1]

Q1	[Solution]
(a)	$x_2 = x_1 + \left(\frac{3}{2}\right)^1 = a + \frac{3}{2}$ $x_3 = x_2 + \left(\frac{3}{2}\right)^2 = a + \frac{3}{2} + \left(\frac{3}{2}\right)^2 = a + \frac{15}{4}$ $x_4 = x_3 + \left(\frac{3}{2}\right)^3 = a + \frac{3}{2} + \left(\frac{3}{2}\right)^2 + \left(\frac{3}{2}\right)^3 = a + \frac{57}{8}$
(b)	$x_2 = x_1 + \left(\frac{3}{2}\right)^1 = a + \left(\frac{3}{2}\right)^1$ $x_3 = x_2 + \left(\frac{3}{2}\right)^2 = a + \left(\frac{3}{2}\right)^1 + \left(\frac{3}{2}\right)^2$ $x_4 = x_3 + \left(\frac{3}{2}\right)^3 = a + \left(\frac{3}{2}\right)^1 + \left(\frac{3}{2}\right)^2 + \left(\frac{3}{2}\right)^3$ … $x_n = a + \left(\frac{3}{2}\right)^1 + \left(\frac{3}{2}\right)^2 + \ldots + \left(\frac{3}{2}\right)^{n-1}$ $= a + \frac{\frac{3}{2}\left(\left(\frac{3}{2}\right)^{n-1} - 1\right)}{\frac{3}{2} - 1}$ $= a + 3\left(\left(\frac{3}{2}\right)^{n-1} - 1\right)$ $= (a-3) + 3\left(\frac{3}{2}\right)^{n-1}$

(c)	As $n \to \infty$, $\left(\frac{3}{2}\right)^{n-1} \to \infty$, i.e., $x_n \to \infty$ Thus the **sequence** $\{x_n\}$ diverges.

2 With reference to the origin O, the points A and B have position vectors $\begin{pmatrix}1\\7\\2\end{pmatrix}$ and $\begin{pmatrix}4\\-2\\2\end{pmatrix}$ respectively. The point C divides the line segment AB such that $AC : CB = 2 : 1$.

(a) Find the position vector of C. [1]

The point D is the mid-point of OB and the point E has coordinates $(1, k, 0)$.

(b) Find the value of k such that C, D and E are collinear. [2]

(c) Find the possible values of k such that the length of projection of $\overrightarrow{OE}$ onto $\overrightarrow{AB}$ is $\sqrt{10}$. [3]

Q2	**Solution**
(a)	Using the ratio theorem, $\overrightarrow{OC} = \dfrac{2\begin{pmatrix}4\\-2\\3\end{pmatrix}+\begin{pmatrix}1\\7\\2\end{pmatrix}}{3} = \begin{pmatrix}3\\1\\2\end{pmatrix}$
(b)	$\overrightarrow{OD} = \dfrac{1}{2}\begin{pmatrix}4\\-2\\2\end{pmatrix} = \begin{pmatrix}2\\-1\\1\end{pmatrix}$ C, D and E are collinear $\Rightarrow \overrightarrow{CE} = t\ \overrightarrow{CD}$ for some $t \in \mathbb{R}$ $\overrightarrow{CD} = \begin{pmatrix}-1\\-2\\-1\end{pmatrix}$ and $\overrightarrow{CE} = \begin{pmatrix}-2\\k-1\\-2\end{pmatrix} = 2\begin{pmatrix}-1\\\dfrac{k-1}{2}\\-1\end{pmatrix}$ We have $\dfrac{k-1}{2} = -2 \quad \Rightarrow \quad k = -3$
(c)	Length of projection $= \dfrac{\left\lvert\overrightarrow{OE} \bullet \overrightarrow{AB}\right\rvert}{\left\lvert\overrightarrow{AB}\right\rvert}$ $= \dfrac{\left\lvert\begin{pmatrix}1\\k\\0\end{pmatrix} \bullet 3\begin{pmatrix}1\\-3\\0\end{pmatrix}\right\rvert}{3\sqrt{1+9}} = \dfrac{\lvert 1-3k\rvert}{\sqrt{10}}$ Given: $\dfrac{\lvert 1-3k\rvert}{\sqrt{10}} = \sqrt{10}$ $\Rightarrow \quad 1-3k = 10 \quad$ or $\quad 1-3k = -10$ Hence $\quad k = -3 \quad$ or $\quad k = \dfrac{11}{3}$

3 **Do not use a calculator in answering this question.**

The complex number z satisfies the equation

$$z^2(-2+\mathrm{i})+4\mathrm{i}z+q=0, \text{ where } q \text{ is a real number.}$$

It is given that one root is of the form $p + p\mathrm{i}$, where p is a non-zero real number.

(a) Find the values of p and q. [4]

(b) Find the other root of the equation. [3]

Q3	Solution
(a)	Subst $z = p + p\mathrm{i}$ into $z^2(-2+\mathrm{i})+4\mathrm{i}z+q=0$, $(p+p\mathrm{i})^2(-2+\mathrm{i})+4\mathrm{i}(p+p\mathrm{i})+q=0$ $(p^2+2p^2\mathrm{i}-p^2)(-2+\mathrm{i})+4p\mathrm{i}-4p+q=0$ $-4p^2\mathrm{i}-2p^2+4p\mathrm{i}-4p+q=0$ $-2p^2-4p+q-4p^2\mathrm{i}+4p\mathrm{i}=0$ Comparing real and imaginary parts: $-2p^2-4p+q=0$ --- (1) $-4p^2+4p=0$ --- (2) From (2), $4p(-p+1)=0$ $\Rightarrow p=0$ (rej $\because p\neq 0$) or $p=1$ Subst into (1): $-2(1)^2-4(1)+q=0 \Rightarrow q=6$
(b)	Let r be the other root of the equation. $z^2(-2+\mathrm{i})+4\mathrm{i}z+6=(-2+\mathrm{i})(z-(1+\mathrm{i}))(z-r)$ Comparing **coefficient of z^0**: $6=(-2+\mathrm{i})(-(1+\mathrm{i}))(-r)$ $r=-\dfrac{6}{(2-\mathrm{i})(1+\mathrm{i})}$ $=-\dfrac{6}{(2+2\mathrm{i}-\mathrm{i}+1)}$ $=-\dfrac{6}{3+\mathrm{i}}\times\dfrac{3-\mathrm{i}}{3-\mathrm{i}}$ $=-\dfrac{6(3-\mathrm{i})}{10}=-\dfrac{9}{5}+\dfrac{3}{5}\mathrm{i}$ OR Let $z^2(-2+\mathrm{i})+4\mathrm{i}z+6=(z-(1+\mathrm{i}))((-2+\mathrm{i})z-r)$ Find r by comparing **coefficient of z^0**: Then the other root is $z=\dfrac{r}{-2+\mathrm{i}}=\dfrac{r}{-2+\mathrm{i}}\times\dfrac{-2-\mathrm{i}}{-2-\mathrm{i}}=\ldots$

4 A curve C has parametric equations

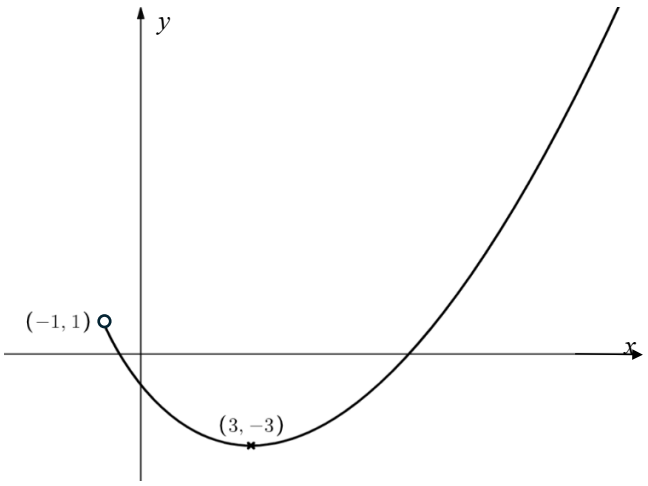
$$x = t^2 - 1, \quad y = t^3 - 3t^2 + 1, \quad \text{for } t > 0.$$

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of t, and hence find the coordinates of the stationary point of C. [4]

(b) Sketch the graph of C for $t > 0$. You do not need to find the intersections of the curve with the axes. [2]

(c) Find the equation of the tangent to curve C at the point P where $t = 4$. [2]

(d) Using a non-graphical method, determine if the tangent to curve C at point P will meet C again. [3]

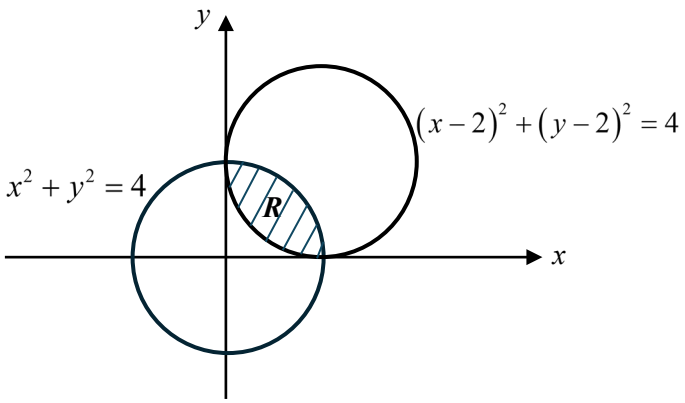
Q4	**Solution**
(a)	$x = t^2 - 1, \quad y = t^3 - 3t^2 + 1 \ , \ t > 0$ $\frac{\mathrm{d}x}{\mathrm{d}t} = 2t, \quad \frac{\mathrm{d}y}{\mathrm{d}t} = 3t^2 - 6t$ $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3t^2 - 6t}{2t} = \frac{3}{2}(t - 2)$ At stationary points, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \ \Rightarrow \ t = 2$ Coordinates of the stationary point of C is $\left(2^2 - 1, \quad 2^3 - 3(2)^2 + 1\right) = (3, -3)$
(b)	
(c)	When $t = 4$, $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3}{2}(4 - 2) = 3, \quad x = 15, \quad y = 17$ Equation of the tangent to C at P is $y - 17 = 3(x - 15)$, i.e., $y = 3x - 28$

Q4	Solution
(d)	C: $x = t^2 - 1$, $y = t^3 - 3t^2 + 1$ Tangent at P meets curve C: $t^3 - 3t^2 + 1 = 3\left(t^2 - 1\right) - 28$ $\Rightarrow \quad t^3 - 6t^2 + 32 = 0$ $\Rightarrow (t-4)^2(t+2) = 0$ [This can be obtained either by GC, or noting that $t = 4$ is where the tangent touches (and is tangent to) C, hence there is a repeated factor.] $\Rightarrow t = 4$ (which is point P) or $t = -2$. Since we are given that $t > 0$, the tangent at P does **not** meet curve C again.

5 **(a)** Use the substitution $x = 2 + 2\sin\theta$ to find the exact value of $\int_0^2 \sqrt{4-(x-2)^2}\ \mathrm{d}x$. [6]

(b) The diagram below shows the two circles with equations $x^2 + y^2 = 4$ and $(x-2)^2 + (y-2)^2 = 4$. The shaded region R is bounded by the two curves. Find the exact volume of the solid formed when R is rotated through 2π radians about the x-axis. [5]

[The volume of a sphere with radius r is given by $\frac{4}{3}\pi r^3$.]



Q5	Solution
(a)	$x - 2 = 2\sin\theta \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}\theta} = 2\cos\theta$ When $x = 0,\ \sin\theta = -1 \Rightarrow \theta = -\frac{\pi}{2}$ When $x = 2,\ \sin\theta = 0 \Rightarrow \theta = 0$ $\int_0^2 \sqrt{4-(x-2)^2}\ \mathrm{d}x = \int_{-\frac{\pi}{2}}^0 \sqrt{4-(2\sin\theta)^2}\ (2\cos\theta)\mathrm{d}\theta$ $= \int_{-\frac{\pi}{2}}^0 2\cos\theta(2)\sqrt{1-\sin^2\theta}\ \mathrm{d}\theta$ $= 4\int_{-\frac{\pi}{2}}^0 \cos^2\theta\ \mathrm{d}\theta$ $= 2\int_{-\frac{\pi}{2}}^0 1+\cos 2\theta\ \mathrm{d}\theta$ $= 2\left[\theta + \frac{\sin 2\theta}{2}\right]_{-\frac{\pi}{2}}^0$ $= 2\left[0 - \left(-\frac{\pi}{2} - 0\right)\right]$ $= \pi$

(b)

$(x-2)^2+(y-2)^2=4 \Rightarrow y=2\pm\sqrt{4-(x-2)^2}$

Required volume

= Volume of hemisphere $-\ \pi\int_0^2 y^2\ \mathrm{d}x$

$=\frac{1}{2}\left[\frac{4}{3}\pi(2)^3\right]\ -\ \pi\int_0^2\left(2-\sqrt{4-(x-2)^2}\right)^2\ \mathrm{d}x$

$=\ \frac{16}{3}\pi\ -\ \pi\int_0^2\left(4-4\sqrt{4-(x-2)^2}+4-(x-2)^2\right)\mathrm{d}x$

$=\ \frac{16}{3}\pi\ -\ \pi\int_0^2\left(8-(x-2)^2-4\sqrt{4-(x-2)^2}\right)\mathrm{d}x$

$=\ \frac{16}{3}\pi\ -\ \pi\left[8x-\frac{(x-2)^3}{3}\right]_0^2+4\pi(\pi)$

$=\ 4\pi^2+\frac{16}{3}\pi\ -\ \pi\left[(16-0)-\left(0+\frac{8}{3}\right)\right]$

$=4\pi^2-8\pi$ or $4\pi(\pi-2)$

Section B: Probability and Statistics [60 marks]

6 Anand plays a game at a funfair. In each game, he is given 3 tokens which he tosses one at a time at a target area filled with red cups and black cups. It may be assumed that the probability of him tossing a token into a red cup is a constant 0.7.

Points are given for each game based on the number of tokens that are tossed into red cups. The points system is shown in the table below.

Number of tokens in red cups	Score
0	0 points
1 or 2	1 point
3	3 points

The number of points that Anand scores in one game is denoted by S.

(a) Show that $\mathrm{E}(S) = 1.659$. [3]

(b) Find $\mathrm{Var}(S)$. [1]

Anand wants to win a stuff toy that needs 100 points for redemption. He plays n games and is allowed to accumulate his points over multiple games.

(c) State the least number of games he has to play to have a chance of winning the stuff toy. [1]

(d) Find the smallest integer n for which there is at least 90% chance that Anand can redeem that stuff toy after playing n games. [4]

Q6 **Solution**

(a)

Let X be the number of tokens (out of 3) that land in a red cup.

$X \sim \mathrm{B}(3, 0.7)$

The probability distribution of S is

s	0	1	3
$\mathrm{P}(S = s)$	$\mathrm{P}(X = 0)$ $= 0.3^3$ $= 0.027$	$1 - 0.027 - 0.343$ $= 0.63$	$\mathrm{P}(X = 3)$ $= 0.7^3$ $= 0.343$

$\mathrm{E}(S) = 1(0.63) + 3(0.343) = 1.659$ (Shown)

(b)

$\mathrm{E}(S^2) = 1^2(0.63) + 3^2(0.343) = 3.717$

$\mathrm{Var}(S) = \mathrm{E}(S^2) - [\mathrm{E}(S)]^2$

$= 3.717 - 1.659^2$

$= 0.964719$ (exact)

(c)

To have a chance of winning the stuff toy (total score of 100 points), he needs to play at least 34 games.

Q6	Solution
(d)	Let $T = S_1 + S_2 + ... + S_n$, the total score of n games. For a total score of 100 points, need at least 34 games. Since $n > 30$ is sufficiently large, **by Central Limit Theorem**, $T \sim \mathrm{N}(1.659n,\ 0.964719n)$ approximately Given: $\mathrm{P}(T \geq 100) \geq 0.9$ From GC, n | $\mathrm{P}(T \geq 35)$ 66 | 0.8829 < 0.9 67 | 0.9173 > 0.9 Thus least n is 67.

7 Beng, Charlie and 6 other friends bought tickets to their favourite band's concert. These tickets are for 8 consecutive seats in a row.

(a) Find the number of ways the 8 friends can seat themselves if Beng is seated next to Charlie on Charlie's left. [1]

To support their band, everyone agrees to dress in either blue or silver outfits.

(b) Find the number of ways the 8 friends can choose the colour of their outfits if there must be at least two outfits of each colour. [3]

After the concert, the 8 friends go to a café for dinner. The café has only 2 available tables: a round table with 5 seats and a round table with 3 seats.

(c) Find the number of ways they can seat themselves if Beng and Charlie are seated at the same table. [3]

Q7	**Solution**
(a)	Required number of ways $= 7! = 5040$
(b)	Required number of ways = total no. of ways – no. of ways all in blue or silver – no. of ways exactly one person is in blue or silver $= 2^8 - 2 - {}^8C_1 \times 2$ $= 238$ **<u>Alternative method</u>** No. of ways $= {}^8C_2 + {}^8C_3 + {}^8C_4 + {}^8C_5 + {}^8C_6 = 238$
(c)	**Case 1:** Beng and Charlie seated on round table of 3 No. of ways $= {}^6C_1 \times (3-1)! \times (5-1)! = 6 \times 2! \times 4! = 288$ **Case 2:** Beng and Charlie seated on round table of 5 No. of ways $= {}^6C_3 \times (5-1)! \times (3-1)! = 20 \times 4! \times 2! = 960$ Total required number of ways $= 288 + 960 = 1248$ **<u>Alternative method</u>** Required number of ways = total no. of ways with no restrictions – no. of ways Beng and Charlie are seated at different tables $= {}^8C_5 \times (3-1)! \times (5-1)! \quad - {}^6C_4 \times (3-1)! \times (5-1)! \times 2$ $= 2688 - 1440$ $= 1248$

8 In this question, you should state the parameters of any distributions you use.

In a game, a contestant must complete two tasks A and B in the shortest time possible. The times taken to complete task A have the distribution N(m, 1.25) and the times taken to complete task B have the distribution N(7, 1). The unit for time is minutes, and the times taken to complete the tasks are independent.

It is given that 3.1% of the contestants will take at least 22.8 minutes to complete both tasks.

(a) Show that $m = 13.0$, correct to 1 decimal place. [3]

(b) To win a prize, a contestant must take no longer than k mins to complete both tasks. The game master only wants to give a prize to the fastest 10% of the contestants who complete both tasks. Find the value of k. [2]

(c) Find the probability that the total time taken by 3 randomly chosen contestants to complete both tasks is less than 62 minutes. [3]

(d) Find the probability that the time taken by a randomly chosen contestant to complete task A is greater than twice the time taken by another contestant to complete task B. [3]

Q8	**Solution**
(a)	Let X and Y be the time taken, in minutes, to complete task A and B by a contestant respectively. $X \sim \text{N}(m, 1.25)$ $Y \sim \text{N}(7, 1)$ Let $T = X + Y$, total time taken (in mins) by a contestant to complete both tasks A and B. $T \sim \text{N}(m + 7,\ 2.25)$ Given: $\text{P}(X + Y \geq 22.8) = 0.031$ $\text{P}\left(Z \geq \dfrac{22.8 - (m+7)}{\sqrt{2.25}}\right) = 0.031$ $\dfrac{15.8 - m}{1.5} = 1.866296$ $m = 13.0$
(b)	$T = X + Y \sim \text{N}(20,\ 2.25)$ Given: $\text{P}(X + Y \leq k) = 0.1$ From GC, $\text{P}(X + Y \leq 18.0777) = 0.1$ Thus $k = 18.1$ (to 3 s.f.)

Q8	Solution
(c)	$\mathrm{E}(T_1 + T_2 + T_3) = 3(13) + 3(7) = 20$ $\mathrm{Var}(T_1 + T_2 + T_3) = 3(1.25) + 3(1) = 6.75$ $T_1 + T_2 + T_3 \sim \mathrm{N}(60,\ 6.75)$ Required probability $= \mathrm{P}(T_1 + T_2 + T_3 < 62)$ $= 0.779$ (3 s.f.)
(d)	$\mathrm{E}(X - 2Y) = 13 - 2(7) = -1$ $\mathrm{Var}(X - 2Y) = 1.25 + 2^2(1) = 5.25$ $X - 2Y \sim \mathrm{N}(-1,\ 5.25)$ Required probability $= \mathrm{P}(X - 2Y > 0)$ $= 0.331$ (3 s.f.)

9 A production manager compiles data on rice crop yield, y tons/ha, at different amounts of rainfall, x cm. The data is tabulated as shown below.

x	60	80	100	120	140	160	180	200
y	2.1	2.8	3.5	4.2	4.6	4.9	5.0	4.8

The value of the product moment correlation coefficient is 0.93244 correct to 5 significant figures.

(a) What does the correlation coefficient suggest about the relationship between rice crop yield and amount of rainfall? Justify your answer. [1]

(b) Sketch a scatter diagram of the data and explain why the answer in part **(a)** should be changed. [2]

(c) Suggest a reason, in context, why a linear model $y = a + bx$, where a and b are constants, is not an appropriate model for the relationship between x and y. [1]

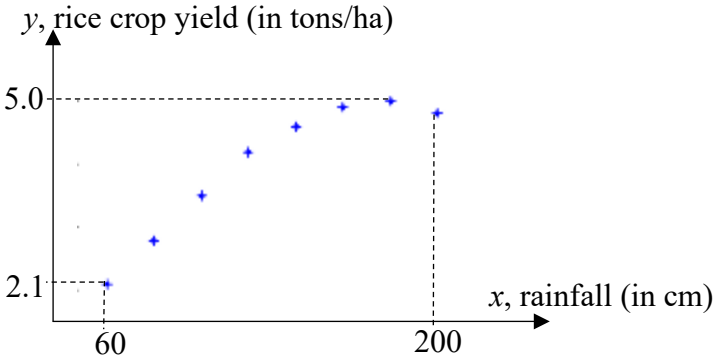
The production manager thinks that the data should be modelled by the quadratic equation $y = c + d(x - H)^2$, where H is a suitable constant. The product moment correlation coefficient between $(x - H)^2$ and y is denoted by r. The following table gives the values of r for some possible values of H.

H	182	183	184	185
r	–0.997383	–0.997390		–0.997293

(d) Calculate the value of r for $H = 184$, giving your answer correct to 6 decimal places. [1]

(e) Use the table and your answer to part **(d)** to suggest with a reason which of 182, 183, 184, 185 is the most appropriate value for H. Using this value for H, calculate the values of the constants c and d. [2]

(f) Give an interpretation, in context, of the values of H and c. [2]

Q9	[Solution]
(a)	Since the correlation coefficient of 0.93244 is close to 1, it suggests that rice crop yield and amount of rainfall has a strong positive linear correlation.
(b)	 The scatter diagram shows that the data points increase at a decreasing rate and then later decreases and not increasing at a constant rate. Thus the diagram does not support the linear correlation suggested in (a).
(c)	As x, the amount of rainfall increases, the rice crop yield is unlikely to increase because too much water can severely damage crops. • Nutrient loss: Flooding washes away nitrogen and other essential nutrients. • Yield reduction: Prolonged waterlogging leads to stunted growth and lower grain production. • Disease outbreaks: Fungal infections like sheath blight thrive in humid, waterlogged conditions. • Mechanical damage: Strong currents or deep floods can physically damage rice plants.
(d)	From GC, when $H = 184$, $r = -0.997359$ (6 d.p)
(e)	The most appropriate value for H is 183 since its correlation coefficient $r = -0.997390$ is the r-value closest to -1. From GC, the regression line of y on $(x-183)^2$ is $y = 4.93927 - 0.00019388(x-183)^2$ Thus $c = 4.94$ (3 s.f) and $d = -0.000194$ (3 s.f.)
(f)	$y = 4.93927 - 0.00019388(x-183)^2$ H represents the optimal amount of rainfall (in cm) that gives the maximum yield of rice crop, c tons/ha.

10 In a large library, past records show that

- 80% of books borrowed are Fiction,
- 15% are Non-fiction, and
- 5% are Reference books.

Assume that each borrower can only borrow one book at a time, and each borrower borrowed books independently of one other.

(a) Three borrowers are randomly selected.

(i) Explain what is meant by randomly selected, in the context of the question. [1]

(ii) Find the probability that all three borrowed books of different types. [1]

(iii) Find the probability that at least one borrowed a Reference book. [2]

(iv) Given that at least one borrowed a Reference book, find the probability that none borrowed Fiction books. [3]

(b) A librarian selects a random sample of 10 borrowers.

(i) State, in context, one assumption needed for the number of borrowers in the sample who borrowed Fiction books to be well modelled by a binomial distribution. [1]

Assume now that the number of borrowers who borrowed Fiction books in the sample follows a binomial distribution.

(ii) Find the probability that the number of borrowers who borrowed Fiction books is between 5 and 9 inclusive. [2]

(iii) The librarian repeatedly selects 50 samples of 10 borrowers. Find the expected number of samples containing between 5 and 9 borrowers inclusive who borrowed Fiction books. [1]

(c) Find the least number of borrowers that the librarian must select in his sample such that he is at least 90% confident that there is more than one borrower in the sample who borrowed a Reference book. [3]

Q10	**[Solutions]**
(a) (i)	It means that **each** borrower has an equal chance of being **selected** and they are **selected** independently.
(ii)	Required probability = $0.8 \times 0.15 \times 0.05 \times 3!$ $= 0.036$
(iii)	Required probability = 1 – P(none borrowed a Reference book) $= 1 - (0.95)^3$ $= 0.142625$ (exact decimal) **Alternative solution** Required probability = ${}^3C_1(0.05)(0.95)^2 + {}^3C_2(0.05)^2(0.95) + (0.05)^3 = 0.142625$

Q10	[Solutions]
(iv)	P(none borrowed Fiction book | at least one borrowed a Reference book) $= \dfrac{\text{P(1 Ref, 2 Non Fict)+ P(2 Ref, 1 Non fict)+ P(3 Ref)}}{\text{P(at least 1 Ref)}}$ $= \dfrac{{}^3C_1(0.05)(0.15)^2 + {}^3C_2(0.05)^2(0.15) + (0.05)^3}{0.142625}$ $= \dfrac{0.04625}{0.142625}$ $= 0.0324 \quad (\text{or } \dfrac{37}{1141})$
(b) (i)	The **probability** that a borrower borrowing a Fiction book **remains constant** for **all** borrowers.
(ii)	Let X be the number of borrowers (out of 10) who borrowed a Fiction book. $X \sim \text{B}(10, 0.8)$ Required probability $= \text{P}(5 \le X \le 9)$ $= \text{P}(X \le 9) - \text{P}(X \le 4)$ $= 0.88626$ (5 s.f.) $= 0.886$ (3 s.f.)
(iii)	Expected number of samples $= 50 \times 0.88626$ $= 44.3$ (3 s.f.)
(c)	Let Y be the number of borrowers (out of n borrowers) who borrowed a Reference book. $Y \sim \text{B}(n, 0.05)$ Given: $\text{P}(Y > 1) \ge 0.9$ $\Rightarrow \quad 1 - \text{P}(Y \le 1) \ge 0.9$ $\Rightarrow \quad \text{P}(Y \le 1) \le 0.1$ $\text{P}(Y \le 1) \le 0.1$ Using GC n | $\text{P}(Y \le 1)$ 76 | 0.1014 > 0.1 77 | 0.0973 < 0.1 Least number in the sample = 77

11 A tea manufacturer claims that its teabags have a mean mass of 3.2 grams. A quality control officer takes a random sample of 50 teabags and measures their masses.

The masses, x grams, are summarised as follows:

$$n = 50 \qquad \sum x = 156 \qquad \sum (x - \bar{x})^2 = 7$$

(a) Find unbiased estimates of the population mean and variance of the masses of the teabags. [2]

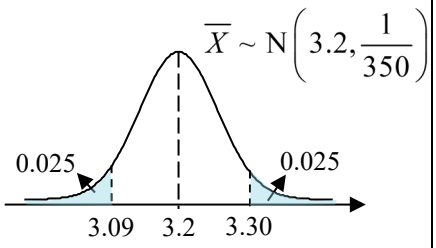
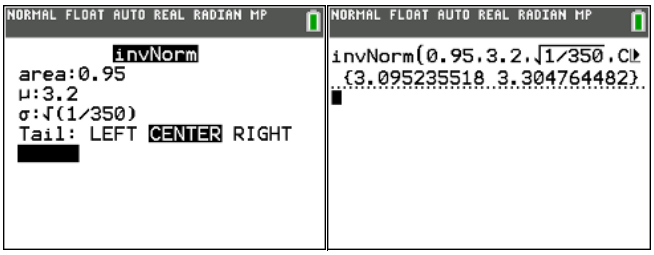
(b) The quality control officer wishes to test, at the 5% level of significance, whether the mean mass of the teabags has differed from 3.2 grams.

(i) State the hypotheses and find the critical region for this test. You should define any parameters you use. [3]

(ii) State the conclusion of the test in the context of the question. [2]

(c) The officer's supervisor suspects that the mean mass of the teabags is less than 3.2 grams. Given that the supervisor concluded that his suspicion is justified at $\alpha\%$ level of significance, find the range of the possible values of α. [3]

Q11	**Solution**
(a)	Unbiased estimate of the population mean, $\bar{x} = \frac{156}{50} = 3.12$ (exact value) Unbiased estimate of the population variance, $s^2 = \frac{1}{50-1}\sum (x - \bar{x})^2 = \frac{7}{49} = \frac{1}{7}$

Q11	Solution
(b) **(i)**	Let μ be the **population** mean mass of a teabag, in grams. H_0: $\mu = 3.2$ H_1: $\mu \neq 3.2$ Perform a 2-tail test at 5% level of significance Under H_0, since $n = 50$ is large, by Central Limit Theorem, $\overline{X} \sim N\left(3.2, \frac{1}{7(50)}\right)$ approximately $\overline{X} \sim N\left(3.2, \frac{1}{350}\right)$  Reject H_0 if $\overline{x} \leq 3.09$ or $\overline{x} \geq 3.30$ (3 s.f.) Critical region is $\{\overline{x} \in \mathbb{R}^+ : \overline{x} \leq 3.09 \text{ or } \overline{x} \geq 3.30\}$ GC Screenshots: 
(b) **(ii)**	Since $\overline{x} = 3.12$ does not fall into the critical region, we do not reject H_0. **Alternative method** From GC, p-value $=0.13448 > 0.05 \Rightarrow$ do not reject H_0. Thus, there is insufficient evidence at the 5% level of significance for the quality control officer to conclude that the mean mass of the teabags differs from 3.2 grams.
(c)	H_0: $\mu = 3.2$ H_1: $\mu < 3.2$ Under H_0, using the same sample data, p-value $= \frac{0.13448}{2} = 0.06724$ Suspicion is justified at $\alpha\%$ level of significance $\Rightarrow$ reject H_0 $\Rightarrow$ $p - \text{value} = 0.06724 \leq \frac{\alpha}{100}$ $\Rightarrow$ $\alpha \geq 6.72$ (3 s.f)